MUHAMMAD AFFAN NUR IMAN BIN AZMAN

CUSTOM IC LAYOUT ENGINEER | ANALOG / MIXED-SIGNAL | GRADUATE ENGINEER

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PROFESSIONAL SUMMARY

Mechatronic Engineering graduate from Universiti Sains Malaysia with practical experience in 28nm analog and 40nm standard-cell layout using Cadence Virtuoso and Siemens Calibre DRC/LVS. Completed op-amp and bandgap-reference layout exercises with DRC-clean and LVS-matched results. Experienced in floorplanning, matched-device placement, routing, physical verification debugging and SKILL productivity utilities. Brings cross-disciplinary robotics and embedded-systems experience, with a self-directed approach to continuous technical development and AI-assisted learning.

CUSTOM IC LAYOUT EXPERIENCE

Custom Layout Engineer Trainee (OJT) | Vchip Technology, Penang | 22 Jun 2026 - 31 Oct 2026

- **28nm analog layout:** Completed operational-amplifier and bandgap-reference layout exercises; achieved DRC-clean and LVS-matched results using Calibre.
- **Bandgap and LDO:** Developing a separate bandgap/LDO top-level layout, with emphasis on floorplanning, sensitive-device placement, supply routing and inter-block connections.
- **Analog matching:** Applied interdigitation, common-centroid principles, device symmetry, consistent device surroundings and critical-net routing for matching-sensitive structures.
- **Physical verification:** Investigated DRC geometry errors and LVS connectivity mismatches by interpreting foundry rules and cross-checking the schematic against the layout.
- **Custom layout productivity:** Created or customised Cadence SKILL utilities for cross-rulers, labels, metal-layer changes and bindkeys to streamline repetitive work.
- **40nm CMOS standard cells:** Laid out single/multi-finger inverters, NAND/NOR, AND/OR, XOR, half adders and full adders.

IC DESIGN TRAINING

IC Design Custom Layout | Penang Skills Development Centre (PSDC) | May - Jun 2026

- **32nm CMOS:** Implemented logic gates and hierarchical adder exercises; performed DRC/LVS using Synopsys Custom Compiler and IC Validator.
- **14nm FinFET:** Digital layout training, including half/full-adder and multi-bit adder exercises.

TECHNICAL SKILLS

EDA tools: Cadence Virtuoso, Siemens Calibre DRC/LVS; Synopsys Custom Compiler and IC Validator (training).

Layout implementation: Floorplanning, matched-device placement, interdigitation, common-centroid concepts, MOS fingers, diffusion sharing, guard rings, well taps, manual routing and power rails.

Circuit and reliability knowledge: Differential pairs, current mirrors, BJT and resistor networks, bandgap/LDO building blocks; WPE, LOD, parasitic RC, EM/IR, latch-up and coupling.

Programming & workflow: Cadence SKILL, Python, C/C++, Linux; AI-assisted technical research, scripting and troubleshooting with independent validation.

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ADDITIONAL ENGINEERING EXPERIENCE & CREDENTIALS

RESEARCH & ROBOTICS ENGINEERING

Research Assistant - Autonomous UAV Systems | Universiti Sains Malaysia | 2025 - 2026

- Integrated Pixhawk-based flight controllers, Raspberry Pi companion computing, telemetry and sensor interfaces; tested control workflows with ArduPilot/PX4 and ROS/Gazebo simulation.
- Performed system-level troubleshooting across hardware, control software and communication links for unmanned aerial platforms.

Unmanned Surface Vehicle Research Support | USM / PETRONAS-related project | 2025 - 2026

- Supported restoration and debugging of a Clearpath Heron M300 after computer/SSD failure, addressing ROS, networking, propulsion and navigation-sensor integration.

Robotics Engineer Intern | Cytron Technologies | Jul - Oct 2024

- Built educational robots and embedded-system demonstrations, developed technical tutorials and conducted a Raspberry Pi AI-vision workshop.

SELECTED PROJECT & ENGINEERING STRENGTHS

- **Final-year project:** Designed and tested a twin-gyrostabiliser for an unmanned surface vehicle using BLDC/servo actuation, IMU measurements and PID control.
- **Multidisciplinary experience:** UAV, UGV/AMR and USV/ROV integration; microcontrollers, companion computing, embedded sensors and technical prototyping.
- **AI-assisted engineering:** Uses AI tools to accelerate structured learning, explore implementation options and prototype scripts, while validating outputs against documentation, simulation and engineering requirements; avoids sharing confidential IP.

EDUCATION & PROFESSIONAL REGISTRATION

Bachelor of Mechatronics Engineering: Universiti Sains Malaysia (USM), Nov 2025.

Graduate Engineer: Registered with Board of Engineers Malaysia (BEM), Jan 2026.

SELECTED CERTIFICATIONS & TRAINING

- **Cadence (Sep 2026):** Virtuoso Layout Design Basics vIC25.1 Exam; Semiconductor 101 v3.0 Exam.
- **PSDC (Jun 2026):** IC Design Custom Layout training in CMOS digital layout and physical verification.
- **ASEM (Jun 2026):** Chip Start 2.0 - Digital and Analog IC Design.
- **ASEM / SkyeChip (Feb 2026):** Custom Layout 1-Day Bootcamp.
- **USM / Lattice Semiconductor (Apr 2025):** FPGA and AI Training.
- **ViTrox College (Dec 2024):** V-ONE Artificial Intelligence Training Program.
- **Additional learning:** CMOS Analog Circuit Design, Introduction to Semiconductor Manufacturing, and Introduction to Generative AI.

LEADERSHIP & LANGUAGES

Leadership: USM ROTU Army Cadet Officer (2022-2024); USM Drone Club Technical Team Committee (2023-2024).

Languages: Malay (native); English (professional working proficiency).